**EIN3241 – Human Factors & Ergonomics**

Homework 2

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Please answer the following questions in a separate document and submit it to Canvas.

# Part A: Multiple Choice Questions (18 pts)

1. Anthropometry literally means:
   a) Human movement
   b) Measurement of humans
   c) Human systems
   d) Human posture

2. The parallel axis theorem is used to:
   a) Convert linear dimensions into angular ones
   b) Calculate MOI about an axis not through the CoM
   c) Estimate ROM based on age and gender
   d) Correlate two anthropometric dimensions

3. Which of the following is a *structural dimension*?
   a) Forward grip reach
   b) Vertical grip reach
   c) Stature
   d) Crawling height in mines

4. Which of the following is a *functional dimension*?
   a) Sitting height
   b) Knuckle height
   c) Arm reach while standing
   d) Hip breadth (sitting)

5. The center of mass (CoM) of a body segment is:
   a) Always located at the proximal joint
   b) The midpoint between the proximal and distal joints
   c) A point where the segment's mass is equally balanced
   d) Equal to the radius of gyration

6. The recommended design approach when reach is the main concern is:
   a) Design for the mean (50th percentile)
   b) Design for the smallest (5th percentile) user
   c) Design for the largest (95th percentile) user
   d) Design for adjustability

7. The moment of inertia (MOI) of a body segment reflects:
   a) Resistance to translational motion
   b) Resistance to rotational motion
   c) The sum of forces acting on the body
   d) The ratio of weight to height

8. A designer wants to ensure clearance in a doorway. Which percentile should they use?
   a) 5th percentile female
   b) 50th percentile overall
   c) 95th percentile male
   d) Average across gender

9. Active Range of Motion (ROM) is generally:
   a) Greater than passive ROM
   b) Equal to passive ROM
   c) Less than passive ROM
   d) Unrelated to passive ROM

10. According to Dempster's body segment parameters, which is true?
    a) Segment mass is expressed as a % of total body mass
    b) Radius of gyration is independent of body size and shape
    c) Center of mass always lies at 50% of segment length
    d) All individuals share identical segment proportions

# Part B: Short Answer (27 pts)

1. Define *static (structural) anthropometry* and *dynamic (functional) anthropometry*. Give one example of each and its use in workplace design. (4 points)
   Static anthropometry: measurement of body dimensions in standard, still postures (e.g., sitting height, stature). Example: designing chair height using popliteal height.

   Dynamic anthropometry: measurement of dimensions during movement or activity (e.g., forward reach, overhead reach). Example: designing control panel reach distances.

2. List three factors that affect anthropometric data. Briefly explain their impact. (6 points)
   1) Age – stature increases until early adulthood, then decreases with age.
   2) Gender – men generally larger in most dimensions; women larger in hip/thigh area.
   3) Posture/clothing – slumping decreases stature, clothing/shoes increase external dimensions.
   4) Others: e.g., ethnicity...

3. What is the Morant technique? When might it be used? (2 points)
   A method using perpendicular grids to project body landmarks for measurement. Use-

ful for capturing accurate anthropometric data in research or when direct caliper measurement is impractical.

4. Describe various anthropometric design principles. Provide an example for each design type. (10 points)
   1) Design for the mean (50th percentile). Example: desk height in classrooms.
   2) Design for one extreme (small user, 5th percentile). Example: reach controls in cockpits.
   3) Design for the other extreme (large user, 95th percentile). Example: doorway clearance.
   4) Design for adjustability. Example: adjustable office chairs, car seats.
   5) Design for various sizes. Example: The design of clothing to fit people from diverse statures.

5. A worker has a stature of 177 cm. Using correlations, estimate whether their arm reach is likely to be above or below average. Briefly explain why correlations between body dimensions matter in anthropometry. (3 points)
   Since stature and arm reach are positively correlated, a worker with above-average stature (177 cm) is likely to have above-average arm reach. Correlations allow estimation of related body dimensions when only one measure (like stature) is available, improving design predictions.

6. Explain the concept of radius of gyration. How is it different from center of mass? (2 points)
   The radius of gyration ($k$) is the distance from the axis of rotation at which the entire mass of a body segment could be assumed to be concentrated without changing its moment of inertia (MOI). It reflects how mass is distributed relative to an axis.
   The center of mass (CoM), on the other hand, is the point where the segment's mass is equally balanced in all directions.

   So, the key difference is that CoM locates the balance point of mass while radius of gyration describes the distribution of mass around an axis for rotational calculations.

# Part C: Exercises (55 pts)

1. You are designing a control panel that requires both **reach** and **clearance**. Which dimensions would you select for each, which percentiles would you use, and how would you balance conflicting needs? Provide sketches or a written justification. (10 points)
   **Dimensions to select**:
   *Reach (functional):* forward grip reach (standing/sitting), vertical grip reach, lateral reach, arm reach envelope based on upper + lower arm link lengths; preferred vs. maximum reach arcs.
   *Clearance (structural):* shoulder breadth (standing), elbow-to-elbow breadth (sitting), hip breadth (sitting) for passage/side clearances, popliteal height & thigh clearance (under-panel/knee space), sitting/standing eye height (sight lines), hand breadth/thickness

(control spacing), head clearance if overhangs exist.

**Percentiles**:
*Reach:* design to the 5th percentile female so the smallest users can access primary controls without leaning/twisting.
*Clearance:* design to the 95th percentile male (plus allowances for clothing/PPE) so the largest users fit without contact or constraint.

**Balancing conflicts:** One is when 5th-percentile reach collides with 95th-percentile clearance:
(1) *Adjustability*: Provide panel height/tilt adjustment or an adjustable seat/stand platform to shift the reach envelope while maintaining knee/thigh/head clearances.
(2) *Zoning*: Place high-frequency, safety-critical controls within the *preferred reach zone* of the 5th-pct female; put low-frequency items in the extended zone.
(3) *Geometry*: Use a slanted panel or stepped tiers to shorten vertical reach without stealing under-panel knee space.
(4) *Spacing*: Size control spacing by 95th-pct hand breadth/thickness to prevent inadvertent activation and accommodate gloves.
(5) *Aisles & edges*: Set side/overhead clearances by 95th-pct shoulder breadth and head clearance with a margin. *Principle*: prioritize safety/clearance for the largest users, then recover small-user reach via adjustability and control zoning.

2. Determine the mass of the torso of the 95th percentile male. The estimated mass of the torso and head combined has an average of 38 kg and a standard deviation of 6.2 kg. The estimated mass of the head, measured separately, has a mean of 8.8 kg with a standard deviation of 1.9 kg. You assume the correlation between head and torso to be 0.1. (15 points)

We define $T$ = torso mass, $H$ = head mass, and $S = T + H$ (torso + head). We are given:

$$\mu_S = 38 \text{ kg}, \quad \sigma_S = 6.2 \text{ kg}, \quad \mu_H = 8.8 \text{ kg}, \quad \sigma_H = 1.9 \text{ kg}, \quad \rho_{T,H} = 0.1.$$

**Step 1: Mean torso mass**

$$\mu_T = \mu_S - \mu_H = 38 - 8.8 = 29.2 \text{ kg}.$$

**Step 2: Standard deviation of torso mass** Since $S = T + H$, the variance identity is

$$\sigma_S^2 = \sigma_T^2 + \sigma_H^2 + 2\,\rho_{T,H}\,\sigma_T\sigma_H.$$

Substituting values:

$$6.2^2 = \sigma_T^2 + 1.9^2 + 2(0.1)(\sigma_T)(1.9).$$

This quadratic in $\sigma_T$ solves to

$$\sigma_T \approx 5.72 \text{ kg}.$$

**Step 3: 95th percentile torso mass** Assuming normality,

$$P_{95} = \mu_T + z_{0.95}\sigma_T = 29.2 + (1.645)(5.72) \approx 38.6 \text{ kg}.$$

3. Applying the anthropometric design principles, find out the dimensions of headband length, ear cup dimensions (length, depth, and width) of the earmuff hearing protection. Data is normal (30 points).

**Figure 1: Earmuff Hearing Protection**

| Headband arc length | Mean | SD |
|---|---|---|
| | 35 | 5 |

| Ear depth (cm) | Mean | SD |
|---|---|---|
| | 2.2 | 1.0 |

| Ear breadth (cm) | Mean | SD |
|---|---|---|
| | 3.0 | 1.3 |

| Ear length (cm) | Mean | SD |
|---|---|---|
| | 7.0 | 2.5 |

To accommodate a large number of users (customers) for the hearing earmuff, we propose designing an adjustable headband and an ear cup based on the principle of the largest user.

## Headband

**Design principle:** Design for the adjustability ($5^{\text{th}}$ percentile female – $95^{\text{th}}$ percentile male)

$5^{\text{th}}$ percentile female $= \mu - Z_{0.05}\,\sigma = 35 - (1.645 \times 5)$ $[Z_{0.05} = 1.645; \text{from normal distribution table}]$

$$= 26.78\,\text{cm}$$

$95^{\text{th}}$ percentile male $= \mu + Z_{0.95}\,\sigma = 35 + (1.645 \times 5)$ $[Z_{0.95} = 1.645; \text{from normal distribution table}]$

$$= 43.23\,\text{cm}$$

$$\text{Adjustable range} = (43.23 - 26.78)\,\text{cm} = 16.45\,\text{cm}$$

Thus, the headband should be 43.23 cm long with an adjustable range of 16.45 cm.

## Ear cup

**Ear protrusion (depth) design principle:** Design for the largest user ($95^{\text{th}}$ percentile male).

Depth of ear cup $= \mu + Z_{0.95}\,\sigma = 2.2 + (1.645 \times 1.0)$ $[Z_{0.95} = 1.645; \text{from normal distribution table}]$

$$= 3.85\,\text{cm}$$

**Ear breadth (width) design principle:** Design for the largest user ($95^{\text{th}}$ percentile male).

Breadth of ear cup $= \mu + Z_{0.95}\,\sigma = 3.0 + (1.645 \times 1.3)$ $[Z_{0.95} = 1.645; \text{from normal distribution table}]$

$$= 5.14\,\text{cm}$$

**Ear length (height) design principle:** Design for the largest user ($95^{\text{th}}$ percentile male).

Length of ear cup $= \mu + Z_{0.95}\,\sigma = 7.0 + (1.645 \times 2.5)$ $[Z_{0.95} = 1.645; \text{from normal distribution table}]$

$$= 11.11\,\text{cm}$$